

Programming Assignment 5

Due at the beginning of your discussion session on
September 29-October 1, 2026

Reading

In addition to the following topics, the quiz syllabus includes any material covered in the lectures:

- Read the quick references on routine names (canvas module) and the guide on repeated code (pages 1-12) (canvas module).
- Read Sections 7 (preamble), 7.1 (except “Hide pointers”), 7.2, 7.5, and 23.4 in Code Complete.

Grading Guidelines

An automatic C (or less) is triggered by:

- Any routine with complexity greater than 4, or by
- Any piece of code that is essentially repeated.

However, a submission that avoids these problems does not necessarily qualify for high quality craftsmanship.

Starting with Programming Assignment 7, an automatic C (or less) will be triggered by improperly named routines.



Programming

As usual, make all the changes discussed in your discussion section. Additionally, you should refactor your code to make sure that it adopts the principles covered in the reading assignments.

Implement `Matrix::toString`. Then, design and implement a main method that takes two matrixes from standard input or a file and prints out the resulting product.

General Considerations

This assignment concludes the matrix multiplication project. In the next assignment, we will turn to a different project.

Your classes may contain as many auxiliary private and package-private methods as you see fit, and additional package-private helper classes may be defined. However, any modification or addition to public classes and methods must be approved by the instructors at the discussion board.

You should write JUnit tests to make sure that your primary methods work as intended. However, we will revisit testing later in the course, so extensive testing is not yet recommended. If you have never used JUnit, the JUnit module on canvas lists resources to get you started. Similarly, your code should have a reasonable number of comments, but documentation is going to be the topic of a future assignment. As a general guideline at this stage of the course, comments and tests should be like those accepted in CSDS 132.

Discussion Guidelines

The discussion will focus on routines, including but not limited to appropriate routine names.

Submission

See instruction submissions on canvas.